

SSL

- * Secure Sockets Layer
- * Developed by Netscape
- * Released 1995
- * Less secure, has known vulnerabilities
- * Slower performance
- * Current usage = ~~obsolete~~ obsolete and no longer recommended
- * Commonly called "SSL certificate"

TLS

- * Transport Layer Security
- * IETF (International Engineering Task Force)
- * 1999
- * More secure with stronger encryption
- * Faster & more efficient
- * Widely used today (TLS 1.2, TLS 1.3)
- * Technically a TLS certificate, but still marketed as an "SSL".

SSL → is deprecated and should not be used.

* SSL/TLS = Encrypted data exchanged b/w a browser & a web server. Verifies the identity of a website, enables HTTPS

SSL = an older, outdated security protocol.

TLS = newer, more secure replacement.

TCP

- * Transmission Control Protocol
- * Connection-oriented
- * Reliable, guarantees data delivery
- * Slower
- * Detects & retransmits lost packets
- * Delivers data in the correct order
- * Uses Acknowledgments
- * Supports flow control
- * Header size 20-60 bytes
- * Usage (web browsing (HTTP/HTTPS))

UDP

- * User Datagram Protocol
- * Connectionless
- * Unreliable, no delivery
- * Faster
- * Detects errors but doesn't retransmit
- * Data may arrive out of order
- * No Acknowledgments
- * No flow control
- * 8 bytes
- * Video streaming, online gaming, DNS

TCP = connection oriented protocol that provides reliable and error-free data transmission b/w devices. (HTTP/s)

UDP = is connectionless protocol that sends data without establishing a connection. faster than TCP but does not guarantee data delivery order.

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